EXPERIMENT 6: IMPLEMENTATION OF DELUGE SCRIPTS USING ZOHO CREATOR

AIM:

To implement Deluge scripts in the WattWise – Household Energy Habit Analyzer application to automate electricity consumption calculations, estimate electricity bills, and calculate average daily energy usage.

PROCEDURE:

1. Opened the WattWise application in Zoho Creator.
2. Selected the Meter Reading Log form and created form workflows using Deluge scripting.
3. Implemented a script to calculate electricity units consumed by subtracting the previous meter reading from the current meter reading.
4. Implemented a script to calculate the estimated electricity bill by multiplying the units consumed by the tariff per unit.
5. Implemented a script to calculate average daily consumption by dividing the units consumed by the number of days.
6. Added conditional checks to handle invalid readings, negative values, and zero days.
7. Tested the scripts using sample meter reading data and checked the calculated results.

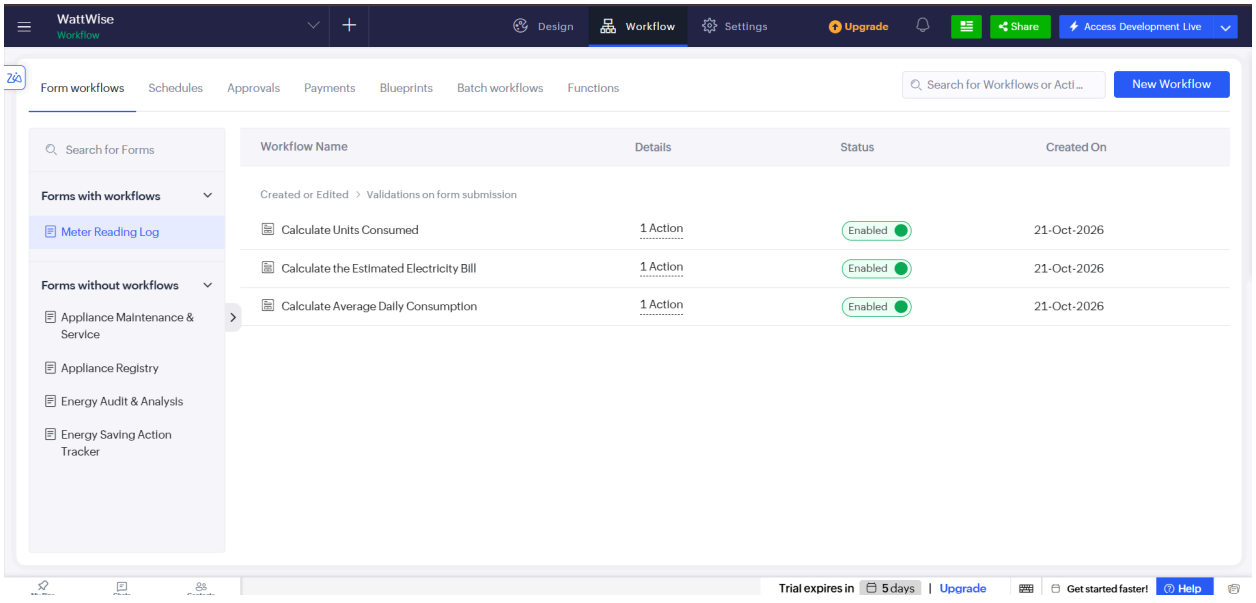
PROGRAMS IMPLEMENTED:

1. **Calculation of Units Consumed:** Calculates electricity consumption using the difference between current and previous meter readings.
2. **Calculation of Estimated Electricity Bill:** Calculates the estimated bill using units consumed and tariff per unit.
3. **Calculation of Average Daily Consumption:** Calculates average daily energy usage using units consumed and the number of days.

SAMPLE INPUT AND EXPECTED OUTPUT:

Description	Input	Expected output
Units consumed	Current reading = 1308 kWh; Previous reading = 1290 kWh	18 kWh
Estimated bill	Units = 18 kWh; Tariff = ₹8/unit	₹144
Average daily consumption	Units = 18 kWh; Days = 4	4.5 kWh/day

OBSERVATION:



Reading ID	Reading Date	Reading Type	Meter Reading (kWh)	Previous Reading (kWh)	Units Consumed (kWh)	Number of Days
3	19-Oct-2026 16:49:12	Daily	1308.00	1290.00	18.00	4
1	17-Oct-2026 13:16:46		1290.00	1250.00	40.00	10
2	15-Oct-2026 16:47:59	Monthly	1290.00	1250.00	40.00	10
5	29-Sep-2026 23:43:34		1800.00	1200.00	600.00	
4	28-Sep-2026 23:39:40	Weekly	1308.00	1290.00	18.00	

Deluge scripting provides a way to automate calculations and validation within the WattWise application. The scripts are designed to calculate electricity consumption, estimate electricity bills, and determine average daily consumption from meter reading

records. Conditional statements are used to identify invalid inputs and prevent calculations with zero days or negative values.

RESULT:

Three Deluge scripts were prepared for the Meter Reading Log form to automate electricity consumption, estimated bill, and average daily consumption calculations. Their outputs can be verified using the specified sample inputs.

CONCLUSION:

The experiment demonstrates how Deluge scripting can automate calculations and validate data in Zoho Creator. These scripts help reduce manual calculation and improve the consistency of household energy records.